

★ WEEK 13 · PART 2 · PRACTICE

The Beta-Binomial, Worked by Hand

Estimate a Win Rate, and Tune the Prior's Strength

Conjugacy makes the update just counting

Part 2's conjugate update, with the arithmetic shown. We estimate a strategy's win rate by adding counts to a Beta prior - watching the estimate shrink from the raw frequency - then see how the prior's strength and the sample size shape the posterior.

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PART 2 · APPLICATIONS

Update by Counting

Case A: estimate a trading strategy's win rate with a Beta-Binomial update, and compare the Bayesian estimate to the naive frequency. Case B: priors as pseudo-counts - weak versus strong - and how the posterior sharpens with data. Then a Claude Code lab.

What Part 2 Told Us

Now we count - with numbers

1

A Beta prior lives on a probability

It's a flexible distribution on the interval from zero to one - perfect for a win rate or a default probability.

2

Conjugacy makes the update arithmetic

A Beta prior plus Binomial data gives a Beta posterior - you just add the wins and losses to the parameters.

3

The prior is pseudo-counts

A Beta's parameters act like trades you've already seen. Their size is how strongly the prior speaks against the data.

Everything here is a calculator exercise: one strategy's record, two priors, two sample sizes.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a Beta prior and some counts.
- Add the counts to get the posterior.
- Then read the mean and the spread.

What you should watch

- How the posterior shrinks toward the prior.
- How prior strength changes the pull.
- How more data sharpens the belief.

Estimate a Strategy's Win Rate

The raw frequency isn't the whole story

A strategy has won 12 of its last 20 trades - a 60% raw win rate. But 20 trades is a thin record. Starting from a mildly skeptical prior, the Beta-Binomial update just adds the wins and losses to the prior's counts, giving a posterior that tempers the raw frequency. We compute it by hand

The inputs

- Prior: Beta(4, 6), a mean of 0.40.
- Data: 12 wins and 8 losses.
- Naive win rate: 12 over 20, or 0.60.

What we'll do by hand

- Add the counts to the prior.
- Compute the posterior mean.
- Compare it to the raw frequency.

The Beta-Binomial Update

Conjugacy: just add the counts

$$\text{Beta}(\alpha, \beta) + \text{data } (s \text{ wins, } f \text{ losses}) \longrightarrow \text{Beta}(\alpha + s, \beta + f) \quad (\text{just add the counts})$$

Why it's so clean

- The Beta prior and Binomial data match.
- The posterior is Beta again - same family.
- No integration, just addition.

The recipe

- Add wins to the first parameter.
- Add losses to the second.
- That's the entire update.

The Posterior

Add the counts, read the mean

$$\text{posterior} = \text{Beta}(4 + 12, 6 + 8) = \text{Beta}(16, 14)$$

$$\text{mean} = \frac{16}{16 + 14} = \frac{16}{30} = \mathbf{0.533}$$

Adding 12 wins and 8 losses to Beta(4,6) gives Beta(16,14). Its mean - the first parameter over their sum - is 0.533, the Bayesian estimate of the win rate.

Bayesian vs Naive

The estimate is pulled toward the prior

$$\underbrace{0.60}_{\text{naive 12/20}} \longrightarrow \underbrace{0.533}_{\text{Bayesian}} \longleftarrow \underbrace{0.40}_{\text{prior}} \quad (\text{shrunk toward the prior on thin data})$$

The gap

- The raw frequency says 60%.
- The Bayesian estimate says 53%.
- The prior of 40% pulled it down.

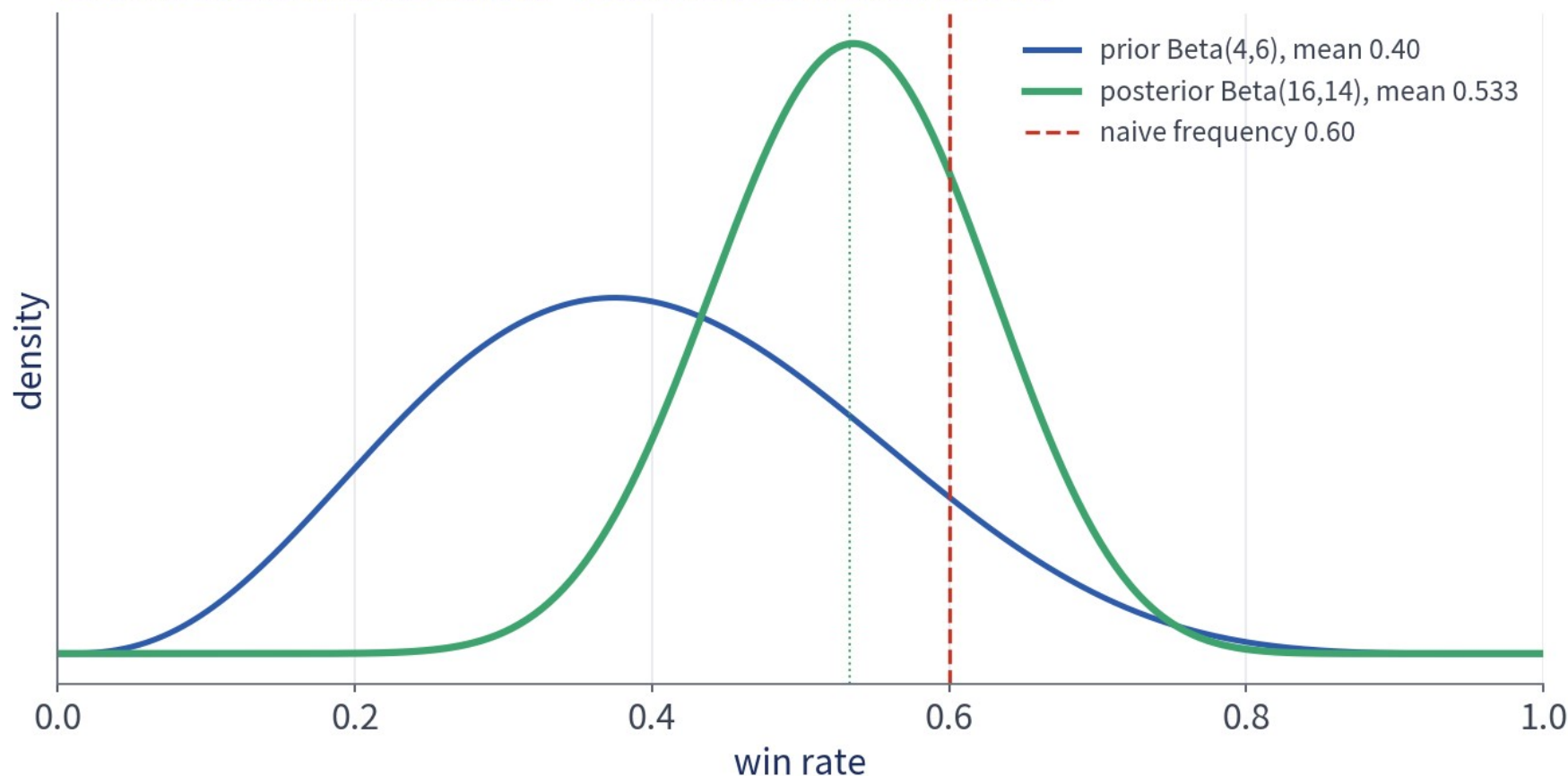
Why it's better

- Twenty trades is a small sample.
- The prior guards against over-reading it.
- The estimate is more robust to noise.

The Update Shifts and Sharpens

The broad prior (blue, centered at 0.40) becomes a tighter posterior (green, at 0.533) once the data arrives. It moves toward the observed frequency of 0.60 but stops short - the prior holds it back, exactly as a thin sample warrants

Case A: the update shifts and sharpens - posterior lands short of the naive 0.60



Shrinkage Is Protection

The prior tames a noisy estimate

The trap

- Taking a 60% win rate at face value.
- Sizing a strategy on 20 lucky trades.
- Chasing a number that won't hold up.

The discipline

- The prior shrinks the estimate toward reality.
- Small samples move belief only so far.
- You get a full distribution, not one number.

How Strong Is the Prior?

Pseudo-counts, and sharpening

A Beta prior's parameters are pseudo-counts: Beta(20,30) is like having already seen 50 trades. So a strong prior resists new data, while a weak one lets the data speak. And as real trades accumulate, the posterior sharpens regardless. We compute both effects by hand

Prior strength

- Weak Beta(1,1): two pseudo-trades.
- Strong Beta(20,30): fifty pseudo-trades.
- Same data, 12 wins and 8 losses.

Sharpening

- Compare 20 trades to 200 trades.
- Same win proportion, more evidence.
- Watch the posterior spread shrink.

Priors as Pseudo-Counts

Prior strength is injected data

$\text{Beta}(\alpha, \beta) \equiv \alpha + \beta$ pseudo-trades seen in advance (prior strength = how much data you inject)

The interpretation

- The parameters are like prior observations.
- Their sum is the prior's effective sample size.
- Bigger sum, stronger belief.

The consequence

- A strong prior needs a lot of data to move.
- A weak prior yields to even a small sample.
- You choose how loudly the prior speaks.

Weak vs Strong Prior

Same data, different pull

weak $\text{Beta}(1, 1) + 12W/8L = \text{Beta}(13, 9)$, mean **0.59** (follows data)
strong $\text{Beta}(20, 30) + 12W/8L = \text{Beta}(32, 38)$, mean **0.46** (resists)

The same 12-8 record gives a posterior mean of 0.59 under the weak prior - close to the data - but only 0.46 under the strong prior, which holds firm near its 0.40 belief.

The Posterior Sharpens

Ten times the data, half the spread

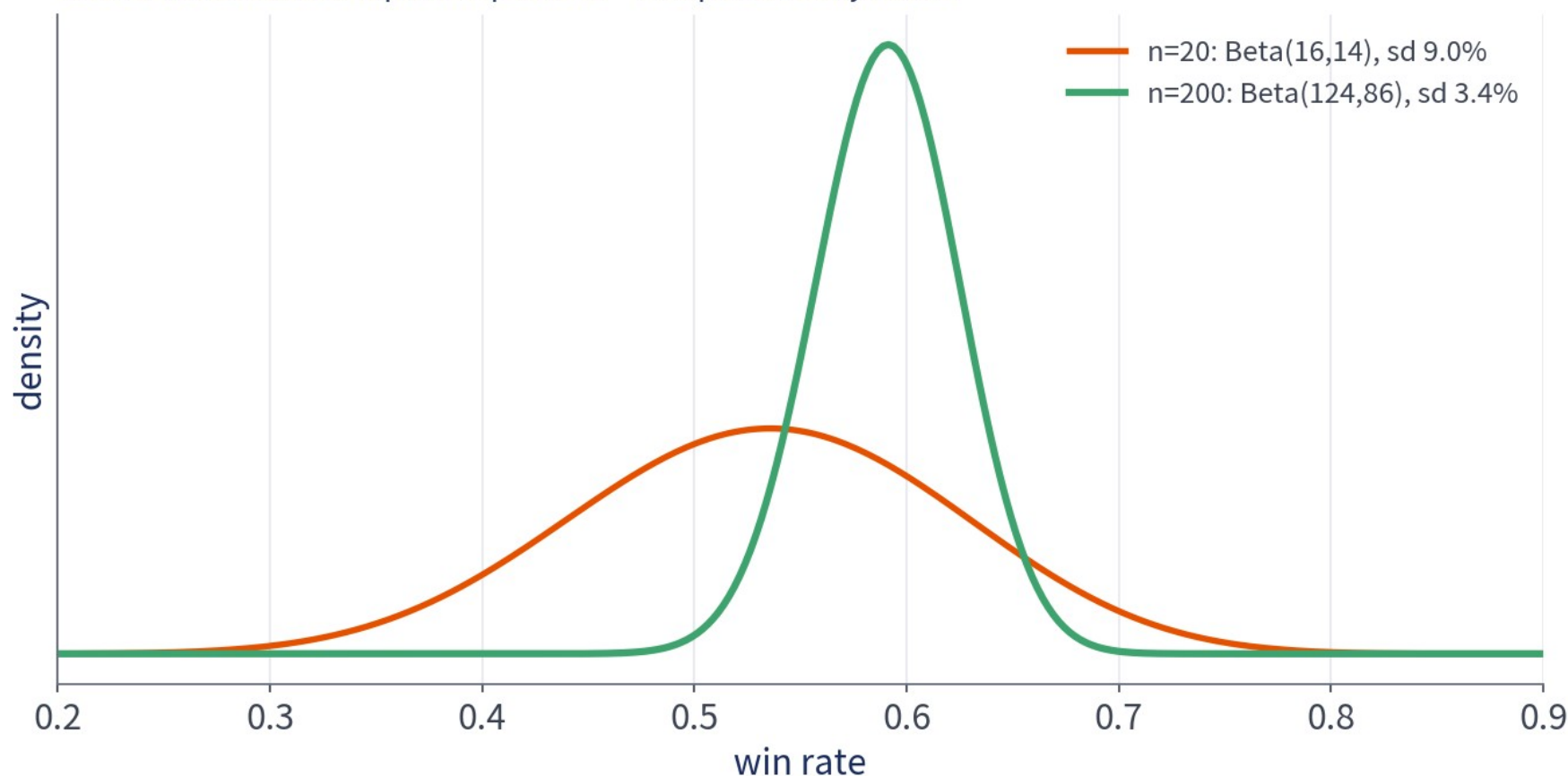
$$\begin{aligned} n = 20 & : \text{Beta}(16, 14), \text{ sd} = 9.0\% \\ n = 200 & : \text{Beta}(124, 86), \text{ sd} = 3.4\% \quad (\text{more data} \Rightarrow \text{sharper posterior}) \end{aligned}$$

With the same win proportion, going from 20 to 200 trades tightens the posterior from a 9.0% spread to 3.4%. More evidence always sharpens the belief.

More Data, Sharper Belief

With 20 trades the posterior (orange) is broad - the win rate could plausibly be anywhere from 0.35 to 0.70. With 200 trades at the same proportion (green), it collapses to a narrow peak. The spread nearly halves; the belief becomes decisive

Case B: more trades sharpen the posterior - the spread nearly halves



Choose the Prior Deliberately

Its strength is a decision

When to lean on the prior

- Little data, or a strong theoretical anchor.
- A strong prior stabilizes the estimate.
- But don't let it drown genuine signal.

When to let data speak

- Plenty of trades, or a weak prior belief.
- The data will dominate anyway.
- The posterior converges on the truth.

Conjugacy Is a Shortcut

When updating is arithmetic

The Beta-Binomial is the cleanest case of a general idea: conjugate priors, where prior and posterior share a family and the update is pure arithmetic. Add wins and losses to a Beta; add observations to a Normal; add counts to a Gamma. The prior's parameters are always pseudo-data, and more real data always sharpens the belief

The update (Case A)

- Beta prior plus counts, Beta posterior.
- Mean shrinks toward the prior.
- Robust on thin data.

The strength (Case B)

- Parameters are pseudo-counts.
- Strong resists, weak yields.
- More data sharpens either way.

Reproduce the Update and the Sharpening

Direct Claude Code to check your arithmetic

GOAL Verify the Beta-Binomial update, the prior-strength effect, and the sharpening.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - the update:

- * from `Beta(4,6)` and 12 wins / 8 losses, print the posterior Beta and its mean; compare to 12/20.

CASE B - strength and sharpening:

- * update `Beta(1,1)` and `Beta(20,30)` with the same data; print both posterior means;
- * print the posterior sd for `n=20` and `n=200`.

Then plot the prior and posterior densities, and the posterior for the two sample sizes.

Match every printed number to the slides.

Check against the slides

- posterior `Beta(16,14)`, mean 0.533.
- naive 0.60, prior 0.40.
- weak 0.59, strong 0.46.
- sd 9.0% -> 3.4%.

Deliverable: `part2_practice.py` whose printed numbers match the update, the strengths, and the sharpening.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you update a Beta prior with Binomial data?

**Add wins and losses
to its two parameters**

Q2 What does a prior's strength represent?

**Pseudo-counts -
trades seen in
advance**

Q3 What happens to the posterior as data grows?

**It sharpens - the
spread shrinks**

All three come from the Beta-Binomial update, pseudo-counts, and the sharpening.

Three Pitfalls to Avoid

Where the Beta-Binomial trips people

Common mistakes

- "The raw win rate is the win rate."
- "A strong prior is always safer."
- "A wide posterior means the model failed."

The reality

- On thin data, shrink it toward a sensible prior.
- Too strong a prior can drown a real signal.
- A wide posterior just means you need more data.

Two Calculations to Remember

Part 2 applications in one screen

1

Update a win rate by counting

Adding a 12-8 record to a $\text{Beta}(4,6)$ prior gives $\text{Beta}(16,14)$ with a mean of 0.533 - shrunk from the naive 0.60 toward the prior's 0.40, because 20 trades is a thin sample. Conjugacy made the update pure arithmetic.

2

The prior's strength is pseudo-counts

The same data gave a posterior mean of 0.59 under a weak $\text{Beta}(1,1)$ but only 0.46 under a strong $\text{Beta}(20,30)$ - fifty pseudo-trades that resist the evidence. And going from 20 to 200 real trades tightened the spread from 9.0% to 3.4%.

3

Conjugacy is the shortcut

Prior and posterior share a family, so updating is just adding counts. The parameters are pseudo-data, prior strength is a deliberate choice, and more real data always sharpens the belief.

Next — Part 3 practice: credible intervals, prediction, and putting Bayesian estimates to work in finance.